What we did since the call of 25 September

An explanation from scratch, with the numbers that back it

Prepared for Giovanni Mantovani

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Abstract

Part I answers the eleven questions Jérôme asked in the call of 25 September. **Part II** is the second test model Jérôme and Marc proposed at the end of September (the consolidated note of 30 September and the email of 1 October): snapshots of a physical field, the ϕ^4 or stochastic Allen–Cahn model. Section 1 explains the method from zero. Each number comes from a run registered in advance, with its output files frozen by a checksum; the few exceptions are flagged. When a prediction we wrote down turned out wrong, the text says so (in *italics*): a falsified prediction is a result, not a failure.

In one line. The paper's claim. For data that form a Markov chain, the score a diffusion model needs can be computed *exactly*, by belief propagation, from a small learned one-step model (a "kernel"); learned from noisy data alone by EM, this beats neural denoisers trained on the same data, because it uses the chain structure that the networks must discover.

1 The method from zero

The data. Sequences $a_1, \ldots, a_L$ that form a *Markov chain*: given a_i, the next value a_{i+1} does not depend on anything earlier. The law is then fixed by the *transition kernel* $K(a' \mid a)$, the density of the next value given the current one (plus the law of the first site, which for a stationary chain follows from K).

What a diffusion model needs. It corrupts the data at a level t, $x = e^{-t}a + \sqrt{\Delta_t}\, z$ with $\Delta_t = 1 - e^{-2t}$ and z standard Gaussian noise, and learns the *score* $\nabla_x \log p_t(x)$, where p_t is the density of the corrupted x. The score is used to denoise and, run backwards from pure noise (the "reverse process"), to generate. By Tweedie's identity

$$\nabla_x \log p_t(x) \;=\; -\,\frac{x - e^{-t}\,\mathbb{E}[a \mid x]}{\Delta_t},$$

so the score is equivalent to the best denoiser in mean squared error, the posterior mean $\mathbb{E}[a \mid x]$.

Why a chain is special. On a chain the posterior mean can be computed exactly by *belief propagation* (BP), the forward–backward algorithm: one sweep left to right and one right to left, each carrying a "message" (a function over one site's value) through K. We store each message on a grid of 401 points on $[-8, 8]$ ("grid BP"). With the true kernel this gives the reference score, accurate to the grid's quadrature error; we call it the **oracle**.

When the kernel is unknown we learn it from the noisy sequences by *EM* (expectation–maximisation): with the current K, run BP on each noisy sequence to get the posterior of every neighbouring pair (E-step); refit K to those soft pair counts (M-step); repeat. "EM–BP" is this, followed by BP with the learned kernel.

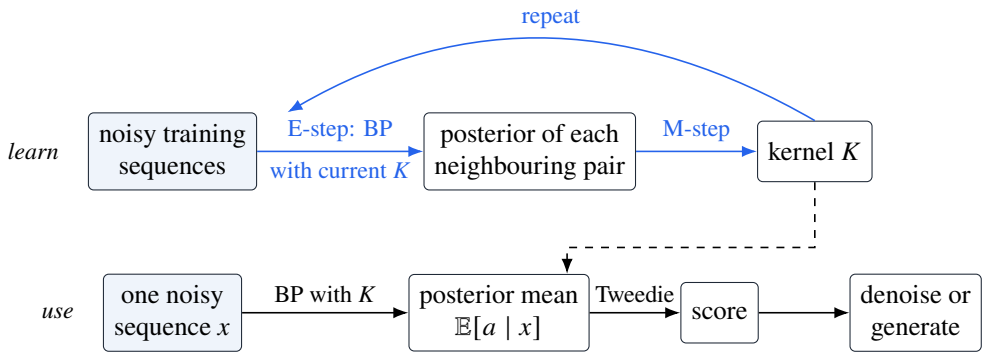


Figure 1: The method. Top: EM learns the kernel from noisy data, with BP inside. Bottom: BP with the learned kernel gives the posterior mean and hence the score. A neural denoiser learns the map from x to the posterior mean (or score) directly instead.

The kernels we fit inside EM–BP (one table for the whole document):

Table 1: Kernel families. The middle three are one model class (a Gaussian step re-weighted by a function of where it lands).

name	what it is	numbers
mixture (the paper's kernel)	next value = $\rho\times$ current + noise drawn from an 8-component Gaussian mixture: the expected next value is *linear* in the current one	25
tilted spline	a Gaussian step re-weighted by a smooth function $w(a')$ of the landing point (log w a spline)	25
Legendre tilt	the same, log w a polynomial	7
site energy	the field's own form (Jérôme's pair energy): a Gaussian "bond" between neighbours plus a quartic potential at each site	5
unrestricted	log K any smooth function of (a, a') (a 10×10 spline); assumes no Gaussian bond	100
MDN	a mixture density network: a small neural net mapping a to a Gaussian mixture for a'	1,516

The competitors. (i) *Neural denoisers* (window CNN, transformer, bidirectional GRU, TCN, U-Net, MLP) trained on the same noisy data. (ii) The **linear baseline** (called the "linear floor" in the reports): the best *linear* denoiser, built from the true covariance. On a linear chain, Gaussian closure of BP (forcing every message to be Gaussian) returns exactly this (the paper's Proposition 1). A method that goes *below* it uses structure beyond second order. In this document "Gaussian" methods always mean this baseline unless said otherwise.

How a method is scored. Its *risk* is a relative RMS error: the square root of the total squared error of its score against the oracle score, divided by the total squared oracle score, summed over test chains, sites and twelve noise levels t from 0.05 to 3. Because it pools sums, the low-noise levels dominate it (the six lowest carry over 90% of the error). Lower is better; 0 is perfect. M is the number of training sequences; a *seed* is an independent repetition with fresh data and initialisation. How long EM runs is chosen on a validation set using the oracle risk; an oracle-free rule (held-out likelihood) is also recorded and gives somewhat larger risks.

How the work is kept honest. Experiments since 25 September (E22, E23, …) have a registry file written before the main run, with predictions and the script that will read them; amendments are dated in the file. Outputs are frozen by checksum (three runs of 26 September were frozen late, on 9 October, after being read). A prediction is read by its own rule: CONFIRMED, FALSIFIED, or UNRESOLVED when the rule cannot decide (for example a model still improving when its budget ran out).

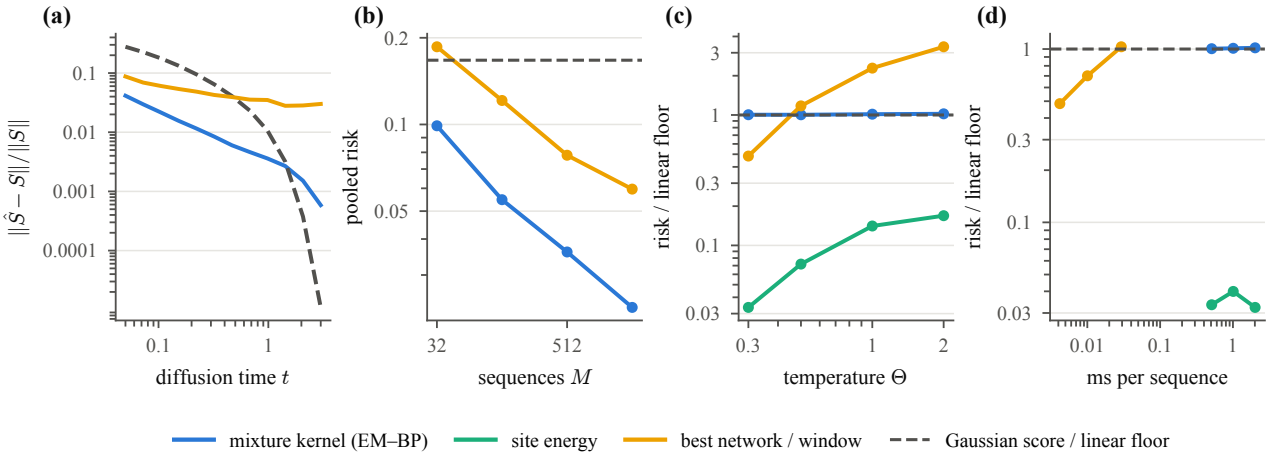


Figure 2: The headline in one picture. (a, b) The paper's test model (Part I): error per noise level at $M = 2048$, and error against M. The learned kernel (blue) beats the best network (yellow) at every level and size, and the Gaussian score (dashed) except at the highest noise levels in (a), $t \gtrsim 2$. (c) The ϕ^4 field (Part II): risk divided by the linear baseline (dashed) across temperature at $M = 2048$. The site energy (green) is far below it, the paper's mixture kernel (blue) sits on it, the window network (yellow) rises above it when hot. (d) The same ratio against inference time (ms per sequence, 8 threads) on three lattice spacings at $\Theta = 0.3$; Section 3.6 explains why the 8-thread timing favours the networks.

2 Part I: Jérôme's eleven questions (call of 25 September)

The test model is the paper's: a linear chain $a_{i+1} = 0.85\, a_i + \varepsilon_i$ with Laplace (sharp-peaked) driving noise ε, $L = 32$ sites, unless a question needed another (stated). The questions fall into five themes.

A. Can BP's messages be summarised by a few numbers? (Q1–Q5). This matters because the 401-point grid is where BP spends its time: a parametric family that BP maps onto itself (a "closed" family) would replace the grid.

- **Q1. Gaussian messages = Gaussian score = Kalman smoother?** Yes on a linear chain: the paper's Proposition 1 proves that BP with Gaussian messages returns the best linear denoiser. Numerically, on a Gaussian chain, the closed form, an independent Kalman (Rauch–Tung–Striebel) smoother and grid BP agree to 1.9×10^{-13}. *The Gaussian-message version run on the 401-point grid was off by up to 0.26, because the grid truncates its widening messages (a registered clause that failed).* [E39]
- **Q2. With Laplace noise, is BP exact with a Laplace-like message family?** No. We tested Jérôme's family (a Gaussian passed once through the kernel) and a Gaussian+Laplace mixture. The family is exact at the first site only; its error (KL divergence to the true message) grows by 7×10^9 to 5×10^{10} from site 1 to site 31. [E22]
- **Q3. What do the messages look like?** Close to that family: at low noise its KL to the true message is 2,272× smaller than a plain Gaussian's and 78× smaller than the 4-parameter mixture's. *We predicted it would do best at high noise; it does worst there, and the mixture overtakes it.* [E22]
- **Q4. Can any finite family close the recursion?** The Gaussian does (on a Gaussian chain). Jérôme's family does not (in 432 of 432 configurations). We have no proof that no other finite family closes; approximately (refit after each step) any family works, at a measured price. [E22]
- **Q5. BP in Fourier space with ~20 modes (Fourier coefficients)?** As a *representation*, partly: on the Laplace chain a Fourier basis on a window adapted to each message needs 33 numbers (KL 10^{-2}) or 47 (10^{-4}) against the grid's 45 and 151, but it reaches 10^{-4} in only 53 of 144 cases and adding modes can make it worse, so a solver could not choose a truncation and know its error. Merely speeding up the transition step on the grid (by FFT) is capped at about 1.15×, because that step is 12–15% of a BP sweep; an FFT version we built is accurate but 5.8× slower at 401 points. The cheap lever turned out to be a coarser grid (Section 3.6). [E23, E24x, E40]

B. Can the kernel be a neural network? (Q6). It can be fitted by EM, and it loses: the MDN is worse than the 25-number mixture kernel in 64 of 64 comparisons (median factor 2.0–2.5), and also worse than the best plain score network (which has 19–25% lower risk). With 61 times more parameters it does 2–2.5 times worse on test data, consistent with the overfitting Jérôme expected. [E25m, E30]

C. Cost (Q7, Q11). It depends on threads and grid size. On the paper's 401-point grid with 8 threads, BP takes 0.52 ms per sequence against 0.042 ms for the transformer (12.5×), and 15× on an H200 GPU. With one thread the networks lose their advantage: BP takes 0.667 ms at 401 points (transformer 0.208, GRU 0.063) and 0.173 ms at 151 points, which is faster than the transformer and TCN and within 15% of the U-Net. Cost grows linearly with the number of sequences and with L; the grid enters like $N_g^{1.2}$, not N_g^2, because elementwise work dominates. [E24x, E47]

D. Sampling without diffusion (Q8). Yes: drawing site 1 and then each next site from the true kernel (ordinary Markov-chain simulation) samples exactly, and is 731× cheaper than the 200-step reverse process that first matches its error (8 threads, 2,048 sequences). With a fitted kernel the samples inherit the kernel's error, the same as the reverse process gives. A Gaussian score generates Gaussian samples: the excess kurtosis of the residual $a_{i+1} - 0.85a_i$ is about 0 against the truth's 1.92 (descriptive, no prediction). [E27, E26]

E. How we evaluate (Q9, Q10). Q9: per noise level, EM–BP beats every network at all 12 levels. It beats the linear baseline (which knows the true covariance) at the 8 (with $M = 128$) or 10 ($M = 2048$) lowest levels; above that the remaining non-Gaussian structure is smaller than the fitted kernel's estimation error, and the crossover moves up with data. [E28] Q10: the information curves were recomputed in cumulant coordinates: fourth-order information falls about 164× more than second-order between $t = 0.05$ and $t = 4$ (a recomputation from frozen E01 files, itself not yet frozen). The paper's panel still plots the shape parameter β labelled "fourth order"; whether to re-plot it is your decision.

In one line. On the paper's test model, EM–BP with a 25-number kernel beats every network at every noise level; exact sampling needs no diffusion; per sequence BP is slower than the networks on many threads, and comparable on one thread with a coarser grid.

3 Part II: the ϕ^4 field

3.1 The model

Think of a one-dimensional elastic string whose height $\phi(x)$ prefers the values +1 or −1, shaken by thermal noise at temperature Θ. Its energy is $H[\phi] = \int \left(\frac{1}{2}\phi'(x)^2 + V(\phi(x))\right) dx$ with the double well $V(\phi) = (\phi^2-1)^2/4$, and its dynamics is the stochastic Allen–Cahn equation $\partial_\tau \phi = \partial_x^2\phi - V'(\phi) + \sqrt{2\Theta}\,\eta$ (η space–time white noise). It settles into the Gibbs law $\propto e^{-H[\phi]/\Theta}$. A snapshot (Figure 3) shows *domains* near +1 and −1 separated by *kinks*; the energy of one kink is $E_{\text{kink}} = 2\sqrt{2}/3 \approx 0.94$. When cold, domains are long and the values have a two-humped histogram.

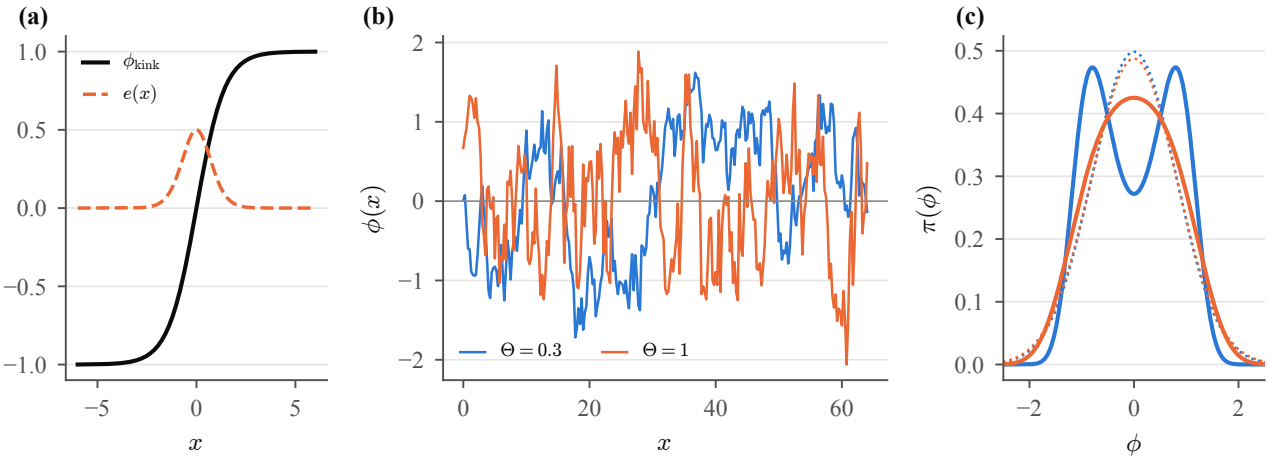


Figure 3: (a) One kink, the field crossing from −1 to +1, and its energy density. (b) Two exact snapshots on a lattice of spacing $h = 0.25$, at $\Theta = 0.3$ (long domains) and $\Theta = 1$. (c) The exact density of the field values (solid) against a Gaussian of the same variance (dotted), $h = 0.25$: two humps when cold.

Why it fits the method exactly. On a lattice of spacing h a snapshot $\phi_1, \ldots, \phi_L$ is *exactly* a Markov chain in the site index, and its kernel is a Gaussian step of variance Θh re-weighted by a function of where it lands (a "tilted walk"; Table 1). The site energy contains the truth exactly; the spline and Legendre tilts are flexible members of the same class. Unlike the paper's linear chain, the expected next value is a non-linear function of the current one (on cold cells it pulls sites towards the two wells), so the mixture kernel, which is linear in this sense, cannot represent it. The *correlation length* ξ (over how many sites correlations decay by a factor e) grows fast as Θ falls.

Six test settings ("cells").

Table 2: The cells, and the risk of two reference methods. The iso cells have h chosen so that the slowest correlation mode decays at the same rate as the base cell's (correlation length about 6 sites).

	base	iso	iso	iso	finer	finest
temperature Θ	0.3	0.5	1	2	0.3	0.3
spacing h	1	0.42	0.23	0.15	0.5	0.25
sites L	32	32	32	32	64	128
linear baseline	0.269	0.115	0.058	0.041	0.162	0.098
each site alone, ignoring its neighbours	0.674	0.657	0.657	0.656	0.714	0.750

The linear baseline measures how much there is to gain beyond a Gaussian model at each cell. It falls with temperature, and also under refinement, where the histogram stays just as two-humped but neighbouring sites become more redundant. [E32]

3.2 Checking the instruments

- **The oracle is exact**: on every cell grid BP at 401 points agrees with a four-times finer grid to 1.3×10^{-12} (checked before any learning). Later, on a Gaussian chain, grid BP matched two independent exact answers to 1.9×10^{-13}. [E32, E39]
- **Gaussian closure by moment matching is not well posed on the field**: its backward messages are nearly flat and pile up at the grid edge. So the Gaussian method is the linear baseline itself. [E32]
- **The paper's information-decay theorem** (Theorem 4 in the current draft: the squared gap to the Gaussian score is a known constant times e^{-8t} at large t) is not reached within the schedule: the measured/predicted ratio at $t = 3$ is 0.85, 0.86, 0.73, 0.94 on the four $L = 32$ cells, and on the $\Theta = 1$ cell it falls over the last step. *Registered as "approaches 1": unresolved.* [E32]

- **Cost** (measured on the paper's chain; BP's cost does not depend on the model): one BP call on 512 sequences of 32 sites needs about 260 MB, five message-sized arrays, linear in everything, 1.8–4.2 times the largest network. [E38]

3.3 Learning the kernel through noise: the main result

Each training sequence is seen once, in one noisy copy at one of the twelve noise levels; no method sees clean data.

- **The 5-number site energy is the best method on every cell at every size** (on average over seeds). At $M = 2048$ its risk is 3–17% of the linear baseline (4–23% with the oracle-free stopping rule) and 43–59% below the spline; it reads the temperature off the noisy data within 10% in 96 of 96 fits. With very little data on the hot cells it still loses to the baseline ($M = 32$); from $M = 128$ it wins everywhere. *We had predicted it would be within 10% of the spline; it is far better.*
- **The paper's own mixture kernel learns nothing beyond the linear baseline** (1.00–1.02 times it at $M = 2048$): a kernel whose expected next value is linear cannot pull sites towards ±1. On non-linear physics the kernel family matters.
- **Networks lose to every tilted-walk kernel**: 4–27 times the spline's risk, and above the linear baseline on the hot cells. The window network chose its radius per cell from $\{2, 4, 8\}$ sites and picked 2, 4, 8 at $h = 1, \frac{1}{2}, \frac{1}{4}$, a constant physical reach (at $h = \frac{1}{4}$ the largest offered). Its absolute risk fell (0.149 to 0.118) but relative to the baseline it rose from 0.55 to 1.20: Jérôme's expectation that structure gains under refinement holds. *We had predicted its absolute risk would grow.* Whether an even wider window closes the gap was not tested.
- **The good kernels are physically right**: samples drawn from the spline and the site energy reproduce the kink density and mean domain length within a few per cent, and the whole distribution of domain lengths within 10% at every quantile, statistically indistinguishable from exact samples (4 seeds). The mixture's samples have about 1.4 times too many kinks.
- **Two doubts removed.** The MDN's gap is not a training-budget artefact: four times the EM budget lowers its risk by only 0.4–4.5% (*we predicted more than 10%*). The potential's barrier height is learnable at high temperature, just slowly: at $\Theta = 2$ its error falls like $M^{-0.54}$ and all 16 seeds are within 20% at $M = 32{,}768$.

Other registered predictions that failed here were thresholds set too tight (E33 P2, E37 P1–P2, E42 P2); none changes the picture. [E33, E34, E37, E41, E42]

In one line. With the right kernel family, exact BP using a kernel learned from noisy data beats the linear baseline and every network by a wide margin on real non-linear physics; with the paper's linear-drift mixture it collapses to the Gaussian answer.

3.4 When the assumptions break

A hidden variable shared by the whole sequence (on a Gaussian test chain with a closed-form answer, not the field). Each sequence is $b\,G + U$, with U the linear chain and $G = \pm 1$ a hidden sign per sequence; the data are no longer Markov. The paper's first-order kernel (it conditions on one previous site) gets stuck: at $b = 2$ four times more data change its risk by +0.2%, and at $M = 2048$ it is 1.57 ($b = 2$) and 1.67 ($b = 1$) times *worse* than the linear baseline, in 16 of 16 seeds. Running BP on pairs (value, sign), so that it also infers each sequence's sign, repairs it: the risk keeps falling and ends 6.3 times below the baseline at $b = 2$. *We predicted the repair would change nothing when $b = 0$; it helped.* We traced that to the reflection symmetry $x \to -x$: fitting the plain kernel on each sequence and its sign-flipped copy is 32–39% better than the plain fit, 16–17% better than the repaired model, and equal in average risk to fitting on twice as many sequences (*the registered equivalence test failed on interval width: seeds vary a lot*). [E35, E35x, E46]

Observing only every s-th site ("stride"; kernels fitted on clean, noise-free chains). A thinned Markov chain is still Markov (its kernel is K applied s times) but no longer a tilted walk. The coarse-grained physics (kink

density, domain length) agrees across strides within 8%. At stride 4 the site energy's and the spline's risks grow about 25 and 14 times (16 seeds). There the MDN is below the site energy in 16 of 16 seeds, but it was still improving at its last checkpoint in 15 of 16 fits, so that registered clause is unresolved; E48 (queued) gives it a far longer budget. The noisy-data version under strides was not run. [E36, E36x]

A chain outside the model class. Replace the Gaussian bond by a sharp Laplace bond ($|a' - a|$ instead of $(a' - a)^2$). The site energy and spline nearly stall at a risk of about 0.10, while the unrestricted kernel keeps improving and ends 1.5 times better at $M = 2048$ (8 of 8 seeds). On two true ϕ^4 cells the unrestricted kernel costs 2.5–5.5 times the site energy, and *that price grows with data, the opposite of our prediction*. So keep the site energy for gradient-flow physics, and use the unrestricted kernel as a misspecification test: if it beats both the site energy and the spline at large M, the Gaussian-bond assumption is wrong. [E43]

Data out of equilibrium. Snapshots taken while the field is still coarsening (domains growing) after a sudden cooling are not Gibbs. Off equilibrium there is no oracle, so the reference is the clean snapshot and the ratios are compressed (a perfect denoiser does not reach zero). Against an empirical Gaussian smoother the site energy's ratio is 0.986 shortly after cooling and 0.950 at equilibrium, below 1 in 8 of 8 seeds at every time. *We predicted the flexible spline would take over off equilibrium; it equals the site energy to three digits.* [E44]

3.5 A new use: missing data with uncertainty

Hide part of a noisy sequence (a random half, or a gap of 8 or 16 of the 32 sites) and draw whole posterior sequences with forward-filtering backward-sampling (FFBS), which gives intervals, not only a mean. The learned site energy and spline match the oracle within 0.4% in error and interval width, with the same coverage (0.89 for a nominal 90% interval). The learned mixture behaves like a Gaussian smoother. A Gaussian smoother with the true covariance is only 1.5–4.6% worse on the cold cell, with 7–9% wider intervals, and within 1% at $\Theta = 1$: *we predicted a visible miscalibration; it narrowly over-covers instead.* [E45]

3.6 Making BP cheaper: the grid

Jérôme called the grid the bottleneck. The 401-point grid had only been compared with finer grids; E47 tried coarser ones.

- With the true ϕ^4 kernel, 101 points are already exact to 2×10^{-11} on every cell (smooth integrands).
- The paper's kernels (Laplace noise, not smooth) need more. At 151 points each kernel family's average risk is within 0.08% of its 401-point value (worst single kernel 0.23%). At 101 points the 8-component mixtures fitted on $M = 2048$ sequences move by 2.2% (worst seed 7.6%; 0.12% at $M = 32$), because with lots of data they learn very narrow components. *We predicted at least 90% of them would be resolved at 101 points; only 22% were (none at $M = 2048$).*
- Time per sequence on one thread: 0.053 ms at 51 points, 0.107 at 101, 0.173 at 151, 0.249 at 201, 0.667 at 401.
- The old "BP is 12–18 times slower than networks" figures used 8 threads, which the networks exploit and BP barely does (0.52 ms on 8 threads against 0.67 on one, at 401 points). At one thread and 151 points BP is faster than the transformer and TCN.

Whether the *learned* ϕ^4 kernels are equally happy on a coarse grid is the part still queued. [E47]

In one line. The practical answer to the bottleneck is a coarser grid (four to six times cheaper), not a new Fourier solver.

3.7 Speciation: when does the denoiser commit to a domain?

Jérôme suggested using the temperature sweep to watch "speciation": as noise is removed, the denoiser must at some point *commit* to the sign of each domain. For each site the exact posterior gives $p = P(\text{clean value} > 0)$; the commitment $|2p - 1|$ runs from 0 (undecided) to 1 (certain). E50 records the noise level t_c at which the site-averaged commitment falls through $\frac{1}{2}$ (in the reverse process, "earlier" means at higher noise), and

compares with a Gaussian "twin" that has the same correlations but no humps. All five registered predictions on commitment held:

- the posterior's confidence is calibrated: the share of signs guessed right equals $(1 + \text{commitment})/2$ to within 0.003;
- colder fields commit earlier: t_c falls from 1.61 at $\Theta = 0.2$ to 0.46 at $\Theta = 2$;
- the exact posterior commits earlier than its Gaussian twin at every temperature, by 0.47 in t when coldest and 0.07 when hottest. At $\Theta = 0.2$ it is half-committed when the noise variance is 24 times the signal's; the twin only at 9 times. So the two humps, not only the correlations, decide when the denoiser commits;
- at $\Theta = 0.2$ ($\xi \approx 19$ sites) t_c grows with the chain length (1.03 at $L = 4$ to 1.72 at $L = 64$) and stops growing once the chain is several domains long (1.750 and 1.754 at $L = 128$ and 256);
- for chains shorter than ξ, t_c moves one-for-one with $\frac{1}{2}\ln(1 + \Lambda)$, Λ the largest eigenvalue of the covariance (slope 0.98): the finite-chain analogue of the speciation time $\frac{1}{2}\ln\Lambda$ of Biroli, Bonnaire, de Bortoli and Mézard (2024).

A second part checked the lattice against the continuum. As $h \to 0$ the physical correlation length converges (it moves by 0.03% from $h = 1/16$ to $1/32$; the $h = 1$ lattice overestimates it by 4–9%). *We predicted that a fit of* $\ln\xi = E_{\text{kink}}/\Theta + \frac{1}{2}\ln\Theta + \ldots$ *on* $\Theta \in [0.1, 0.3]$ *would return the kink energy* $2\sqrt{2}/3 \approx 0.943$ *within 2%; it returns 0.988, and 0.93 instead of* $\frac{1}{2}$. The local slope approaches the theory as Θ falls, with a gap shrinking roughly like Θ^2, so the law looks right but these temperatures are not cold enough to pin its constants. [E50]

4 Where things stand

Running. On the cluster, queued behind a maintenance window (no start before 10 October, 12:00; the scheduler's estimate was about 11 October evening): the learned kernels on a coarse grid (E47b), the network kernel with a long budget under strides (E48), and the network kernel on the out-of-class chain (E49).

Answered by frozen evidence. Jérôme's eleven questions (Q5, Q8 and Q10 with the caveats above), and the note's programme: the four-arm comparison on the field, its physical checks (correlations, kinks, domain lengths, bimodality, generated samples), runtime and memory, the Gaussian/Kalman sanity check, the FFT transition, posterior sampling, clean-data learning, the hidden-variable repair, the subsampling protocol (one clause pending E48), and speciation (E50).

An internal audit (9 October) re-derived the numbers in the registry, the reports and the question ledger from the frozen files. It changed no existing verdict, but it found three runs never frozen (now frozen), gave first verdicts to predictions never read (one falsified, one unresolved), replaced numbers that had been quoted from unfrozen laptop runs, and flagged speciation as not addressed (hence E50). The corrections are applied, each logged in its registry file; this document uses the corrected numbers.

Decisions that are yours.

1. The framing Jérôme suggested on 30 September: state the theory for a general Markov kernel, with the linear chain as the first example and the field as the second; where the field goes (body or appendix) under the 8-page body limit.
2. The information panel (Q10): keep the β curve or re-plot it in cumulant coordinates (already computed); a plain relabel would be wrong by up to 1500×.
3. Choose 5–6 of the seven figures in the figures-first document, plus the summary figure.
4. The reply to Jérôme *and Marc* in the week of 12–16 October; tell Jérôme the Milan date once it is confirmed.

Sources: `experiments/registry/E*.json` (hypotheses, predictions, outcomes, corrections) and the frozen outputs they list; `reports/field-laboratory-2026-10-01.pdf` (full detail); `docs/JEROME_CALL_2026-09-25.md` (the question ledger). Branch `claude/phi4-field-2026-10-01`.